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19. ABSTRACT (Continue on reverse if necessary and identify by block number)

This TOP describes tests for measuring and evaluating the capabilities of airborne laser systems in an airborne environment. It describes test procedures for testing range accuracy, target resolution, target illumination, communications systems, and substitute laser systems.

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U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

AMSTE-RP 702-106
*Test Operations Procedure (TOP) 6-3-166 11 September 1986
AD NO.

LASER SYSTEMS, AIRBORNE

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1. SCOPE. This TOP describes tests for measuring and evaluating the capabilities of airborne laser systems in an airborne environment. It describes test procedures for testing range accuracy, target resolution, target illumination, communication systems, and substitute laser systems.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

ITEM	REQUIREMENT
Avionics maintenance facility	To provide maintenance, repair and calibration for laser systems
Flight test range	Flat level terrain
	Targets of various sizes and material composition

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*This TOP supersedes MTP 6-3-166, dated 14 January 1971.
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Footnote numbers match reference numbers in Appendix C.

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	Standard targets that are spherical and present an optical cross section of at least one square meter
	Instrumented systems for Time Space Position Information (TSPI) and data collection
	Communications networks for range safety, flight control, and target control
	Target turntable to provide the capability to position one tracked vehicle at various angles with relation to the flight path
Meteorological station	To measure and monitor weather conditions along flight profiles and around targets
	To meausure scintillation (rotor halo effect) if any
Data reduction facility	To provide data reduction of all electromagnetically recorded test data
Aviation facilities	An adequate number of the proper type, model and series of aircraft are available for the test period and are in flyable condition
	Aircraft maintenance and parking facilities are available during the test period

2.2 Instrumentation.

DEVICES FOR MEAUSURING	PERMISSIBLE ERROR OF MEASUREMENT*
Aircraft tracking systems	± 1.0 M in Range ± 0.5 degrees in Angels

* The permissible error of measurement for instrumentation is the two-sigma value for normal distribution; thus, the stated errors should not be exceeded in more than one measurement of 20.

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Data collection rate	Commensurate with the accuracy requirements of the test. (Normally 10 samples per second)
Ground based laser instrumentation system	
Laser spot tracker	± 0.5 degrees in Angles
Reflected power density measuring system	± 3.0%
Tape measures and rulers	as stated in TOP 7-3-500

3. REQUIRED TEST CONDITIONS.

3.1 General.

a. Consult USATECOM Regulation 70-24[1] to formulate the test plan.

b. Review the test directive to determine the test objectives.

c. Become familiar with the systems technical and operational characteristics.

d. Obtain and become familiar with the Systems Specification, Test and Evaluation Master Plan (TEMP), Required Operational Capability (ROC) and other related documents.

e. Become familiar with test plans from previous tests on related systems if they exist.

f. Incorporate the proper safety precautions into all test procedures. See TOP 7-3-506[2].

(1) Classification of test data as required.

(2) Range signal security.

(3) ADP accredited to process classified/sensitive information.

(4) Personnel security clearances.

(5) Implementation of OPSEC countermeasures.

(6) Security training for test force and range support personnel.

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3.2 Facilities.

3.2.1 Avionics Maintenance Facility. The actions required to prepare a maintenance facility will vary according to the test location and the extent of contractor support. The following are considered minimum requirements for an avionics maintenance facility and should be obtained prior to beginning a test.

a. General facilities. Approximately 500 sq ft with at least two work benches, two desks, chairs, latrine facilities, 115-volt AC power, and telephone connections.

b. Test equipment, tools, and fixtures. Test equipment and tools required will depend on the system being tested. In many instances the system contractor is tasked to provide the required test equipment and tools for use during testing.

c. Special fixtures such as 28-volt DC power supplies, cabling, and connectors should be obtained.

d. Technical data. System technical data must be obtained and made available in the avionics maintenance facility.

e. Training. Army personnel of the correct MOS must be obtained and trained on operation and maintenance of the system. Normally initial training is accomplished at the manufacturer's plant with on-the-job training conducted by the contractor during testing.

f. Spare parts. All initial spare parts should be obtained and be made available in or near the maintenance facility prior to beginning the test.

3.2.2 Test Ranges.

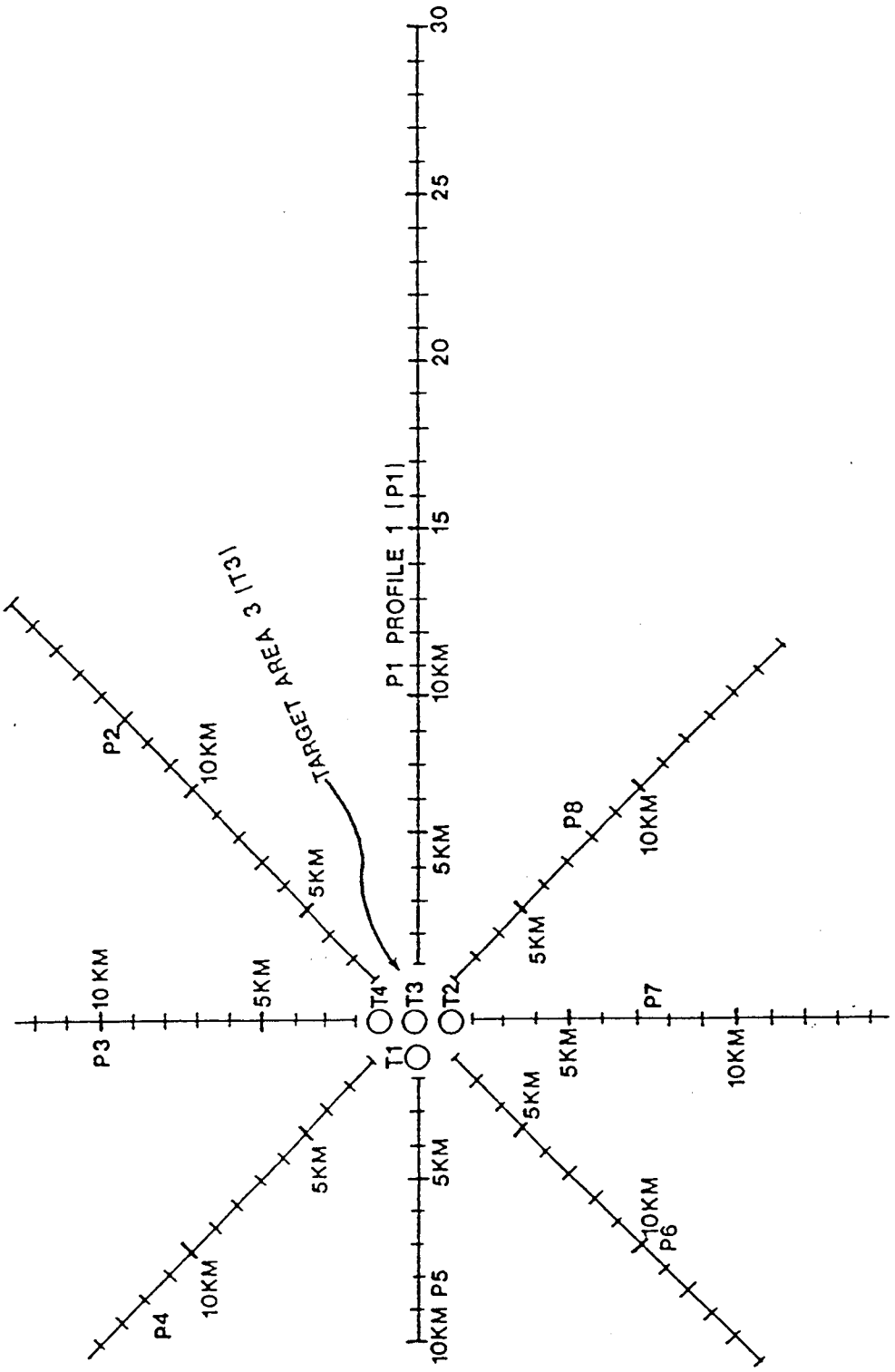
3.2.2.1 Noninstrumented Flight Test Range. Perform the following to set up the test range.

a. Using a topographical map and a road map, locate target location areas that are accessible by ground vehicles. For relative placement see Figure 1.

b. Layout the flight profiles as shown in Figure 1.

c. Have all points and the target areas surveyed and marked with a suitable marker that is easily visible from the air, such as 2 m square white markers with black labels painted on them. The points on the profile should be no more than 0.5 km apart.

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Note: Target locations 1-4 are not shown to scale.

Figure 1. Laser test range.

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d. Measure the range to each target area from each surveyed point, and record the values on the range set-up data collection sheet shown in Appendix B.

3.2.2.2 Instrumented Flight Test Range. Perform the following steps to set-up the test range.

a. Complete paragraph 3.2.2.1a and b.

b. Have all target areas surveyed and market with a suitable marker that is easily visible from the altitudes designated for the on-going test.

c. Record the survey values on the range set-up data collection sheet shown in Appendix B.

3.2.2.3 General Test Range Support.

a. Radiated frequency authorization. Obtain authorization for the frequencies you plan to use.

b. Flight safety requirements. Establish operational safety requirements with the range safety office.

c. Airspace coordination/scheduling. The procedures for scheduling airspace for test missions should be investigated early in the planning phase of a test. The method of scheduling range airspace varies significantly for each test range.

d. Range resources coordination. Coordinate with range resources such as tracking radars, laser trackers, and cinetheodolites to determine the optimum mode of operation for the desired test results. Find out what data are available and how it is obtained.

e. Secure storage facilities. Make arrangements for secure storage facilities at the test range.

f. Target coordination. Ensure all target locations are surveyed. Coordinate with test range personnel for required target types and placement.

g. Petroleum, oils, and lubricants (POL). Ensure aircraft fuel requirements are coordinated with the test range.

h. Test team support. Coordinate with the test range for meals, lodging, and other support for test team members.

i. Range signal security. Determine that telemetry (if used), transmission of test data by FM relay, radio, hardwire circuits, telephone, or ADP networks which process test data requiring protection meet security standards.

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3.2.3 Meteorological station. Brief the station personnel on the conditions which need to be recorded before, during, and after a test mission.

3.2.4 Data reduction facility. Investigate the data reduction capabilities at the test range, and compare them to the data requirements established for the test. If changes are required for the existing software, allow ample time for its devlelopment.

3.3 Instrumentation.

a. Aircraft tracking system. Ensure that the tracking system has the capability for collecting the data required to satisfy the test requirements. As a minimum the tracking system must produce an output data tape containing time tagged range, azimuth, elevation, and tracking status words/bits.

b. Aircraft beacon. Coordinate with the aircraft maintenance facility for installation and verification of proper operation of the beacon.

c. Time code generator. Ensure that the time code generator output is recorded with the video and digital data.

d. Video camera and recorder.

(1) Ensure that an adequate supply of video tapes is available for all test missions.

(2) Ensure that a pre-mission checklist is developed to test the video system.

(3) Ensure that the camera is mounted to collect the required data.

e. Onboard tape recorder.

(1) Ensure that an adequate supply of tapes is available.

(2) Ensure that a pre-mission checklist is developed to test the operation of the tape recorder system.

f. Telemetry transmitter. Ensure that a pre-mission checklist is developed to test the operation of the onboard telemetry system.

g. Ensure that classification instructions have been provided for marking video and tape recorder tapes IAW AR 380-5 when classified information is recorded.

3.4 TEST CONTROLS.

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3.4.1 Data Management. Prepare a data management plan that ensures all test data is properly collected, labeled, and stored during the test.

3.4.2 Test Item Performance Tests. Using the manufacturer's technical manuals, develop test item performance test. This checklist type test should be performed before and after each test mission to verify proper functions.

3.4.3 Quality Control. Check the results of each test mission to ensure no critical data points were missed and to annotate known erroneous data.

3.4.4 Aircraft Instrumentation. Develop an aircraft instrumentation checklist to verify instrumentation systems are in operational order. Complete the checklist before and after each test mission.

3.4.5 Ground Systems. Develop a checklist to verify all ground systems have been set up correctly and are operationally ready to support the test mission.

3.4.6 Cost Tracking. Keep track of test funds expended and remaining for the test program. Expenses should be reviewed on a weekly basis.

4. TEST PROCEDURES. Where possible, two test procedures are presented. One using no instrumentation and one using full instrumentation. They can be used individually or in combination. The determining factors in deciding which test method to use should be the type of test being conducted (developmental or operational) and the accuracy requirements of the data. The performance tests should be performed in the order shown.

4.1 Ranging.

4.1.1 Noninstrumented Flight Test.

a. Method.

(1) Determine the ability of the system to provide accurate and stable distance information in a ranging application.

(a) Use the range set-up procedures in paragraph 3.2.2.1.

(b) Select appropriate targets of various size and material composition. Targets should be placed so as to present known energy reflection. Use one target at a time placed at target location three. Start with the standard target.

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(c) Use profile P1 in Figure 1 for the flight approach pattern to the target. Select the minimum safe altitude that will permit line-of-sight contact with the target. Ground speed should be very slow to allow for greatest accuracy in measurements.

(d) Start at a point approximately two times the maximum rated range for the system. Proceed toward the target, taking range readings at each marker, until a stable accurate range reading can be determined for the target.

(e) Proceed toward the target. Hover over each marker and record range information for the target.

(f) For subsequent passes start at the point determined in step (d) above.

(g) Repeat steps (c) through (f) for each type of target and for as many of the different weather conditions listed under paragraphs 4.6.1 (2) as possible.

(2) Determine the first stable reading range for each target type.

(a) Start at a point approximately two times the maximum rated range for the system. Proceed toward the target, taking range readings at each marker, until a stable accurate range reading can be determined for the target.

(b) Reverse course and fly to the previous marker.

(c) Start inbound until a stable accurate range reading is obtained. Drop a colored marker or sand bag at that point.

(d) Repeat the previous steps at least ten times.

(e) Using a calibrated range measuring device, measure the range from the points that were marked in step (c) to the nearest marker. Record the measured ranges and the marker number they were measured from on the data collection sheet in Appendix B.

(f) Repeat steps (a) through (e) for each type of target and for as many of the different weather conditions listed under paragraph 4.6.1 (2) as possible.

b. Data required.

(1) The data collection sheets shown in Appendix B.

(2) Weather observations.

(3) Pre- and Post-mission checklists for all system.

(4) Video/voice recordings of test missions.

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4.1.2 Instrumented Flight Test.

a. Method.

(1) Determine the ability of the system to provide accurate and stable distance information in a ranging application.

(a) Use the test range described in paragraph 3.2.2.2.

(b) Select appropriate targets of various size and material composition. Targets should be placed so as to present known energy reflection. Use one target at a time placed at target location three. Start with the standard target.

(c) Use profile P1 in Figure 1 for the flight approach pattern to the target. Select the minimum safe altitude that will permit line of sight contact with the target. Ground speed should be very slow to allow for greatest accuracy in measurements.

(d) Start at a point approximately two times maximum rated range for the system. Proceed inbound toward the target, taking range readings at approximately 0.5 Km increments, until a stable accurate range reading is obtained. (Verify range readings with the aircraft tracking system.)

(e) Proceed inbound from the point determined in the previous step. While proceeding inbound, take range measurements on the target at regular intervals. Record the time and range reading of each reading of the data collection sheet.

(f) For subsequent passes start at the point determined in step (d) above.

(g) Repeat steps (d) and (e) for each type of target at different times of day, and weather conditions as shown in paragraph 4.6(a)(2).

(2) Determine the first stable reading range for each target type.

(a) Start at a point approximately two times maximum rated range for the system. Proceed inbound toward the target, taking range readings a approximately 0.5 Km increments, until a stable accurate range reading is obtained. (Verify range readings with the aircraft tracking system.)

(b) Reverse course and fly outbound approximately 0.5 Km.

(c) Start inbound until a stable accurate range reading is obtained.

(d) Repeat steps (b) and (c) at least ten times.

(e) Repeat this procedure for each type of target and for as many different weather conditions listed under paragraph 4.6.(a)(2) as possible.

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b. Data required.

(1) Weather observations.

(2) Pre- and post-mission checklists for all systems.

(3) TSPI, status listing, and aircraft ground track plots with event marks and time.

(4) Onboard instrumentation tape with system status.

(5) Video/voice recordings of test missions.

(6) Data collection sheets.

4.2 Range Resolution.

4.2.1 Noninstrumented Flight Test.

a. Method.

(1) Determine the ability of the system to resolve targets of various size, shape, material composition, and spacing.

(a) Use the range set-up procedures in paragraph 3.2.2.1.

(b) Place one standard target at target location number three.

(c) Place two other standard targets offset from the first in azimuth and range by the distance specified for the laser system for range resolution.

(d) Use the profile labeled P1 in Figure 1 for the flight approach pattern to the targets. Select the minimum safe altitude that will permit line-of-sight contact with the targets.

(e) Start at the range at which the laser beam covers both targets.

(f) Fly inbound on the profile. Record the range at which target resolution is possible.

NOTE: The targets are considered resolved when the range to target two minus the range to target one equals the separation distance of the targets.

(g) At the point where target resolution is achieved, drop a sandbag or colored marker of some type.

(h) Repeat the above procedures at least ten times.

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(i) Measure the distance from the sandbags to the nearest marker. Record these values on the data collection sheet.

(j) Repeat the above procedures for each type of target, using targets of the same type for each test.

b. Data required.

(1) Data collection sheets shown in Appendix B.

(2) Weather observations.

(3) Pre- and post-mission checklists for all systems.

(4) Video/voice recordings of test missions.

4.2.2 Instrumented Flight Test.

a. Method.

(1) Determine the ability of the system to resolve targets of various size, shape, material composition, and spacing.

(a) Use the test range described in paragraph 3.2.2.2.

(b) Place one standard target at target location number three.

(c) Place two other standard targets offset from the first azimuth and range by the distance specified for the laser system for range resolution.

(d) Use profile P1 in Figure 1 for the flight approach pattern to the target. Select the minimum safe altitude that will permit line-of- sight contact with the targets.

(e) Start at the range at which the laser beam covers both targets.

(f) Fly inbound on the profile. Record the range at which target resolution is possible.

NOTE: The targets are considered resolved when the range to target two minus the range to target one equals the separation distance of the targets.

(g) Repeat the above procedures at least ten times.

(h) Repeat the above procedures for each type of target, using targets of the same type for each test.

b. Data required.

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(1) Weather observations.

(2) Pre- and post-mission checklists for all systems.

(3) TSPI, status listings.

(4) Onboard instrumentation tape or telemetered data.

(5) Video/voice recordings of test missions.

(6) Data collection sheets.

4.3. Illuminating/Pinpointing.

4.3.1 Noninstrumented Flight Test.

a. Method.

(1) Determine the ability of the system to provide target illumination for armament systems.

(a) Use the standard target. Place the target on the turntable at target location three.

(b) Use flight profile P1.

(c) Hover at each range marker and illuminate the target. Enter the marker number, time and range on the data sheet for each measure- ment.

(d) Have the other aircraft, equipped with appropriate sensing equipment, fly the other profiles, and determine the range at which sufficient reflected energy is received each time the illuminator aircraft hovers at a range marker. Enter the time, altitude, bearing, and range on the data sheet for each measurement. Confirm proper target acquisition with the illuminator aircraft.

(e) Repeat the above procedure at least ten times.

(f) Replace the standard target with another type target.

(g) Repeat the above procedures for each 45 degree aspect angle of the target.

(h) Repeat the above procedures for each type of target.

(i) Set-up all available targets. Label each target so it can be identified from the air.

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(j) Hover at several range markers and illuminate the targets one at a time.

(k) Have the other aircraft fly the other profiles and try to determine which target is illuminated. Confirm the correct target with the illuminator aircraft.

b. Data required.

(1) Data collection sheets.

(2) Weather observations.

(3) Pre- and post-mission checklists for all systems.

(4) Video/voice recordings of test missions.

4.3.2 Instrumented Flight Test.

a. Method.

(1) Determine the ability of the system to provide target illumination for the armament systems.

(a) Set up a straight line flight profile that terminates at the target. Draw range marks, for the radar controller, at various ranges on the profile.

(b) Set up the laser instrumentation equipment for measuring and recording of the reflected energy, spot size, beam stability, and PRF.

(c) Use the standard target first. Place the target on the target turntable for each test.

(d) Under the aircraft tracking system's guidance, hover at each range mark, and illuminate the target. Record the time and system range.

(e) While the aircraft is illuminating the target, at each range mark, record the parameters listed in step (b).

(f) Rotate the target in 45 degree increments repeating the above procedures for each 45 degree increment. (There is no need to rotate the standard target if a spherical target is used.)

(g) Repeat the above procedures for each target type.

b. Data required.

(1) Weather observations.

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(2) Pre- and post-mission checklists for all systems.

(3) Aircraft tracking system TSPI and status listing.

(4) Onboard instrumentation tape.

(5) Collected data from the laser instrumentation van.

(6) Video/voice recordings.

(7) Data collection sheets.

4.4 Communications.

4.4.1 Noninstrumented Flight Test.

a. Method.

(1) Determine the capability of the system to perform an air-to-ground communications function (voice and/or digital).

(a) Use the range set-up procedures in paragraph 3.2.2.1. Set up the ground based laser communication system on target area three. No other targets or target areas are required.

(b) Fly inbound on the profile while transmitting (digital or voice).

(c) Note the range at which acquisition is made with the ground station and at which station lock is maintained.

(d) Have air and ground operators make notes as to errors in data messages and/or quality of voice transmission, i.e., fading, intermit- tence, interference, loss of transmission, and intelligibility of messages at different ranges. Have aircrews call out the range as they pass the surveyed range markers on the flight profile.

(e) Compare the quality of voice characteristics in step (d) with normal UHF/VHF radio transmissions. Note the differences.

(3) Obtain a measure of beam divergence at different ranges.

(a) This data can be collected in conjunction with steps (1) and (2) above.

(b) While transmitting, vary the angular direction of the laser beam from the direction at which maximum alignment signal is obtained.

(c) Note the angular displacement of the laser beam when communications are lost or become undecipherable.

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(d) Repeat steps (2) and (3) for different ranges.

(4) Determine the security of air-to-ground communications.

(a) This data can be collected in conjunction with step (1) above.

(b) If available, have ground forces equipped with laser receivers try to intercept the transmissions.

(c) If possible, note the range from both the receiving ground station and the transmitting vehicle at which the ground forces can detect the transmission, locate the direction of the transmission, and receive any information in the transmission.

(5) Determine the security of air-to-air communications.

(a) This data can be collected in conjunction with step (2) above.

(b) Fly a third aircraft behind and above the stationary aircraft. The third aircraft should not be allowed to block the laser beam.

(c) Note the range and position from the stationary aircraft that the transmissions were intercepted.

b. Data required.

(1) For paragraph 4.4.1.a:

(a) Range set-up data collection sheets.

(b) Air and ground operators notes.

(c) Recordings of received and transmitted messages.

(d) Operators comments.

(e) Comparison of laser to radio data.

(f) Pre- and post-mission checklists.

(2) For paragraph 4.4.1.b:

(a) Range set-up data collection sheets.

(b) Operators notes and comments.

(c) Comparison and laser and radio data.

(d) Pre- and post-mission checklists.

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(3) For paragraph 4.4.1.c: Same as paragraph 4.4.1.a.

(4) For paragraph 4.4.1.d:

(a) All data for paragraph 4.4.1.a.

(b) Ground forces notes.

(5) For paragraph 4.4.1.e:

(a) All data for paragraph 4.4.1.b.

(b) Interceptor aircraft operator notes.

4.4.2 Instrumented Flight Test.

a. Method.

(1) Perform all steps in paragraph 4.4.1. When aircraft are used, maintain positive control of their positions by means of an aircraft tracking system.

b. Data required.

(1) All data for paragraph 4.4.1.

(2) Aircraft TSPI and tracking status information.

4.5 Substitute Laser Systems.

a. Method.

(1) Determine the ability of a laser system to replace an existing airborne system providing the same function, e.g., laser altimeter versus electronic (radar or CW) altimeter.

(a) For each type of system, equip the aircraft with both the laser system and the existing system it is intended to replace.

(b) Use the TOPs written for the existing system to provide test procedures.

(c) Conduct flight procedures utilizing the existing system for reference readings.

(d) Note variations between the laser system and the existing system.

b. Data required.

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(1) Notes and comments regarding comparisons between systems.

(2) Data collected utilizing test operations procedures for the existing system.

4.6 Effects of Atmospheric Conditions.

a. Method.

(1) Initially conduct performance tests 4.1 through 4.4 during daylight hours with clear stable weather conditions.

(2) Conduct performance tests 4.1 through 4.4 under the following conditions if possible:

(a) Night hours.

(b) At sunrise and sunset.

(c) During poor weather conditions/visibility.

(d) High and low temperatures during the test period.

(e) Frontal passages with changing barometric pressures.

b. Data required.

(1) Weather observations.

(2) Notes and data collection sheets from all test missions.

4.7 Durability.

a. Method.

(1) Conduct durability testing in conjuction with all other testing.

(2) Use the test item checklist to examine the durability characteristics of the system.

b. Data required. Results of all pre- and post-mission checks.

4.8 Maintenance.

a. Method. Use USATECOM Regulation 750-15[3] and TOP 7-3-507[4] to develop test procedures to evaluate the maintenance characteristics of the system.

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b. Data required. As required by USATECOM Regulation 750-15 and 7-3-507.

4.9 Reliability/Availability/Maintainability (RAM).

a. Method. Use TOP 7-3-507, TOP 7-3-508[5], and USATECOM Regulation 750-15 to develop test procedures to evaluate RAM.

b. Data required. As required by TOPs 7-3-507, 7-3-508, and USATECOM Regulation 750-15.

4.10 Electromagnetic Compatibility.

a. Method. Use TOP 7-3-509[6] to determine the electromagnetic compatibility of the transmitter and receiver components.

b. Data required.

c. As required by TOP 7-3-509.

4.11 Draft Technical Manuals.

a. Method. Use MTP 7-3-514[7] to determine the adequacy of draft technical manuals.

b. Data required. As required by MTP 7-3-514.

4.12 Safety Evaluation.

a. Method. Determine the test items safety hazards resulting from transport, operations, and maintenance in accordance with the criteria of TOP 7-3-506.

b. Data required. Record data in accordance with TOP 7-3-506.

4.13 Human Factors Evaluation.

a. Method. Perform a human factors evaluation of the test item using the criteria of TOP 1-2-610.[8]

b. Data required. As required by TOP 1-2-610.

4.14 Training Requirements.

a. Method. Perform a training assessment in accordance with TOP 7-3-501[9].

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b. Data required. As required by TOP 7-3-501.

4.15 Physical Characteristics.

a. Method. Perform a physical characteristics evaluation using the criteria of TOP 7-3-500[10].

b. Data required. As required by TOP 7-3-500.

4.16 Technical Characteristics.

a. Method. Perform technical characteristics test using the appropriate methods in TOP 6-2-165[11].

b. Data required. As required by TOP 6-2-165.

4.17 Aiming and Sighting.

a. Method.

(1) For telescopic systems use the criteria in MTP 10-2-109[12].

(2) For infrared systems use the criteria in TOP 6-2-135[13].

b. Data required. As required by MTP 10-2-109 or TOP 6-2-135.

5. PRESENTATION OF DATA.

5.1 Ranging.

5.1.1 Noninstrumented Flight Test.

a. Separate all test missions by target type.

b. Separate the data from each target type into the following weather catagories:

(1) Visibility greater than 30 Km (clear).

(2) Visibility greater than 15 but less than 30 Km (light haze).

(3) Visibility greater than 5 but less than 15 Km (haze/fog/light rain).

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(4) Visibility less than 5 Km (heavy fog/rain/snow).

c. Separate each of the weather catagories according to time of day:

(1) Day or night.

(2) Sunrise or sunset.

d. Separate each time of day category into test target and standard target data.

e. Separate the test target and standard target data by range markers.

f. Remove all data that were not gathered under the desired test conditions, i.e., measurements on the wrong target.

g. The data that remains is the data available for analysis. The type of analysis used will depend upon the sample size achieved after all data separation is accomplished.

h. Analysis should be accomplished to:

(1) Compare the standard target range readings to the measured distance between the markers and the standard target.

(2) Compare the test target range readings to the measured distance between the markers and the test target.

(3) If more than one laser system was used to collect the same type of data under the same conditions, system-to-system comparisons should be made.

(4) If more than one operator was used to collect the same type of data under the same conditions, operator-to-operator comparisons should be made.

i. Prepare a graph for each weather condition which shows all target types and times of day vs. range. Figure 2 shows a sample graph. Plot the mean range values along with the standard deviation values for each condition.

j. Expected accuracies for this test procedure are 3% of aircraft altitude. For example: if range readings are collected while hovering over marker #6 (3.0 km) at 500 feet AGL (corrected for differences in target and marker elevations), the slant range from the aircraft to the target can be calculated using the pythagorean theorem. The slant range calculated will be $\pm$ 15 ft. (.03 * 500) of the actual slant range. Two assumptions have to be made to arrive at the 3% expected accuracy. They are:

(1) That the pilot can hover over the marker within 3% of his altitude.

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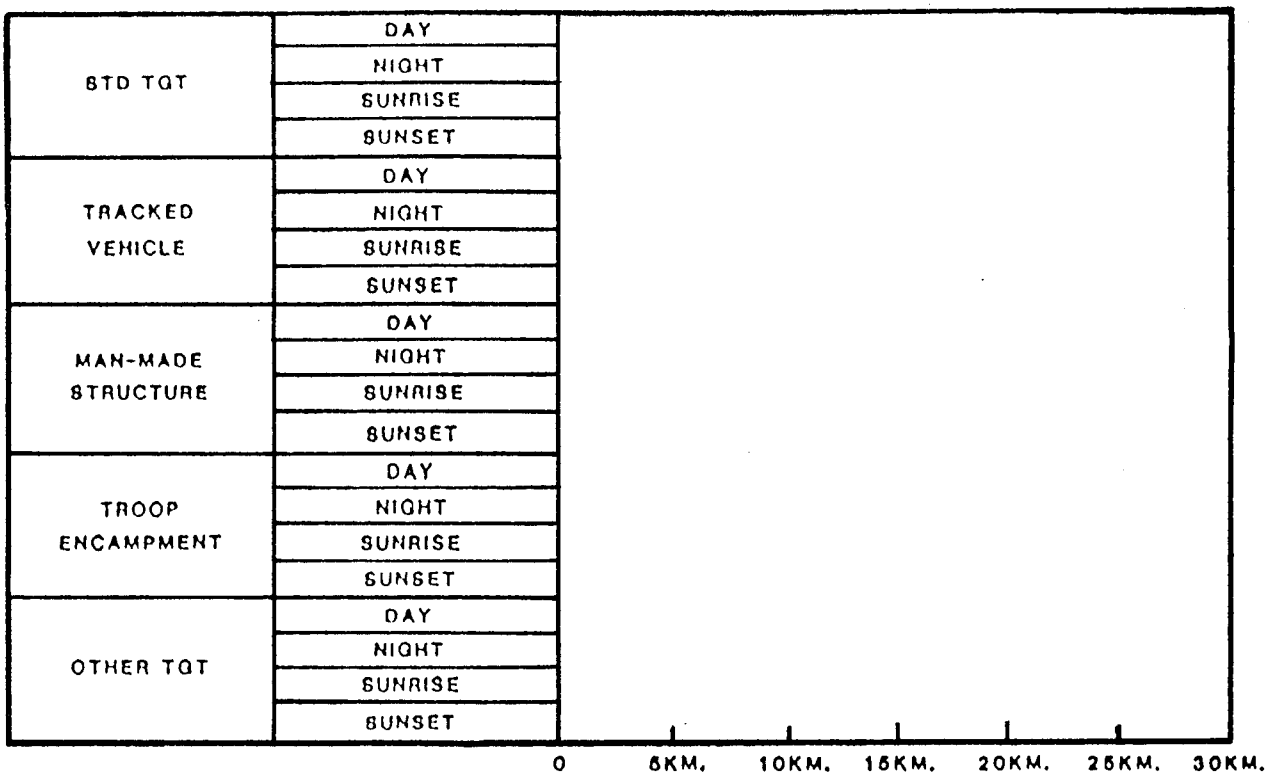


Figure 2. Range Accuracy Data Presentation.

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(2) That a radar altimeter of known accuracy is used to measure the aircraft altitude.

5.1.2 Instrumented Flight Test.

a. Translate and rotate the TSPI data to the standard and test target positions for each test mission.

(1) Produce listings of aircraft azimuth, elevation, and range versus time, referenced to the targets.

(2) Convert all data to engineering units. Use degrees for azimuth and elevation and meters for range.

b. Reduce the aircraft data to engineering units. Produce listings of the range readings and status words versus time.

c. Using the data collection sheets, separate all data by target type, weather conditions, and time of day catagories as explained in paragraph 5.1.1.b through g.

d. Analysis should be accomplished to:

(1) Compare all system range readings to the TSPI range measurements for each target for each test condition.

(2) If more than one laser system was used to collect the same type of data under the same conditions, system-to-system comparisons should be made.

(3) If more than one operator was used to collect the same type of data under the same conditions, operator-to-operator comparisons should be made.

e. Expected accuracy for this test procedure is dependent on the range accuracy of the tracking system and the accuracy of the rotation algorithm used. All range readings should be within $\pm$ 3% of the actual range.

5.2 Range Resolution.

5.2.1 Noninstrumented Flight Test.

a. Separate all test missions by target type, weather, and time of day as described in paragraph 5.1.1.a through c.

b. Separate into time of day categories.

c. Remove all data that were not gathered under the desired test conditions, i.e., measurements on the wrong target.

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d. Analysis should be accomplished to determine the mean and standard deviation for each range at which target resolution was possible.

e. Calculate the system angular resolution for each test condition by dividing the target separation distance by the mean target resolution range.

f. Arrange the data in graphic form as shown in Figure 3.

g. Expected accuracies for this test procedure are 3% of aircraft altitude. See paragraph 5.1.1 for an explanation.

5.2.2 Instrumented Flight Test.

a. Separate all test missions by target type, weather, and time of day as described in paragraph 5.1.1.a through c.

b. Separate into time of day categories.

c. Remove all data that were not gathered under the desired test conditions, i.e., measurements on the wrong target.

d. Merge the aircraft data with the TSPI.

e. Analysis should be accomplished to determine the man and standard deviation for each range at which target resolution was possible.

f. Calculate the system angular resolution for each test condition by dividing the target separation distance by the mean target resolution range.

g. Arrange the data in tabular form as shown in Figure 3.

h. Expected range accuracy for this test is $\pm$ 3% of actual range as described in paragraph 5.1.2.

5.3 Illuminating/Pinpointing.

5.3.1 Noninstrumented Flight Test.

a. Separate the test missions by target type, weather conditions, and time of day as described in paragraph 5.1.1.a through c.

b. Separate the data for each target type into range categories.

c. Separate each range category into target aspect angles.

d. Remove all data that were not gathered under the desired test conditions, i.e., measurements on the wrong target.

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		TARGET SEPARATION		TARGET RESOLUTION RANGE	
		AZIMUTH	DISTANCE	MEAN	STD. DEV.
STD TGT	DAY				
	NIGHT				
	SUNRISE				
	SUNSET				
TRACKED VEHICAL	DAY				
	NIGHT				
	SUNRISE				
	SUNSET				
WHEELED VEHICLE	DAY				
	NIGHT				
	SUNRISE				
	SUNSET				
MAN-MADE STRUCTURE	DAY				
	NIGHT				
	SUNRISE				
	SUNSET				
TROOP ENCAMPMENT	DAY				
	NIGHT				
	SUNRISE				
	SUNSET				

Figure 3. Target Resolution Data Presentation.

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e. Determine the mean range value for each test condition.

f. Prepare a polar plot for each target at each angle, as illuminated from each marker, that shows the target in the center with range toward the outer edge. Plot the slant ranges at the angle of the target at which the receiving aircraft obtained sufficient signal strength to activate armament systems.

g. Figure 4 shows a sample polar plot.

h. Comment on the ability of the other aircraft to identify the correct target when intermixed with multiple targets.

i. The expected accuracy for this test is ± 3% of aircraft altitude as explained in paragraph 5.1.1.

5.3.2 Instrumented Flight Test.

a. Separate the test missions by target type, weather conditions, and time of day as described in paragraph 5.1.1.a through c.

b. Separate the data for each target type into range categories.

c. Separate each range category into target aspect angles.

d. Remove all data that were not gathered under the desired test conditions, i.e., measurements on the wrong target.

e. Determine the mean range value and the standard deviation for each test condition.

f. If testing a pulsed laser, determine the mean PRF of the system.

g. Prepare a polar plot for each target at each angle, as illuminated from each marker, that shows the target in the center with reflected energy toward the outer edge.

h. Prepare a chart that shows beam spot size at the target for each range tested.

i. Comment on beam stability and PRF.

j. Expected range accuracy for this test is ± 3% of actual range as described in paragraph 5.1.2.

5.4 Communications.

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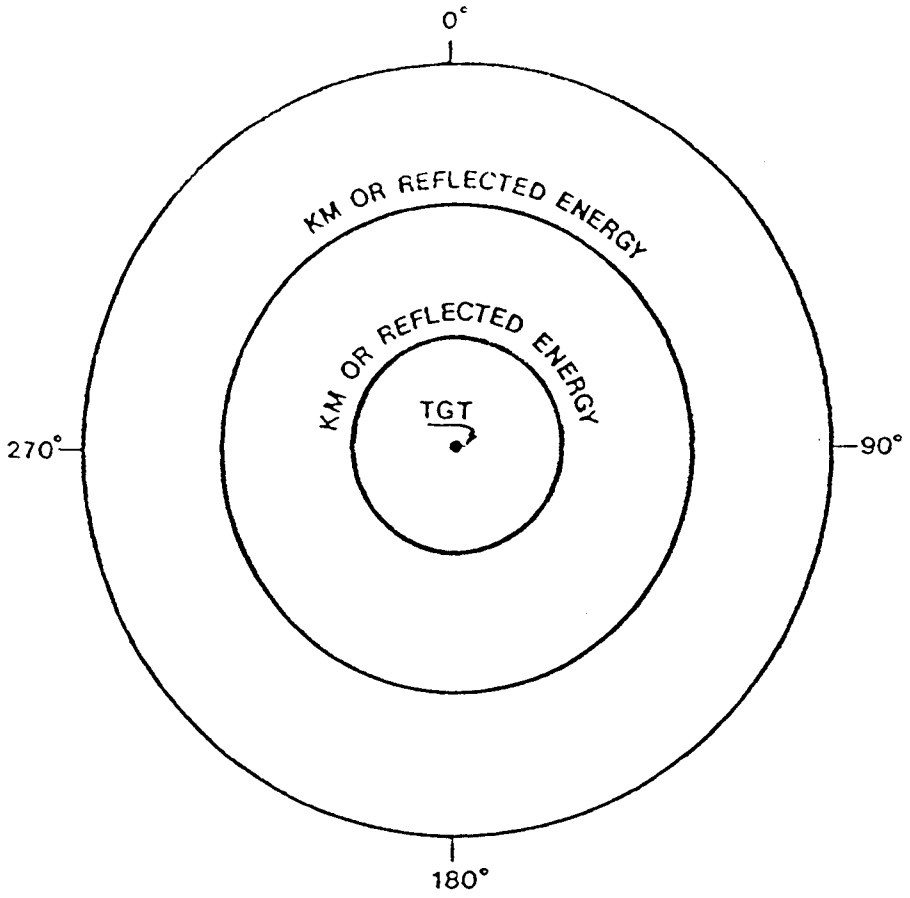


Figure 4. Illuminating/Pinpointing Data Presentation.

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5.4.1 Noninstrumented Flight Test.

a. Separate all test missions into the objectives listed in section 4.4.

b. Separate each objective by weather conditions and time of day.

c. Remove all data that were not gathered under the desired test conditions, i.e., measurements on the wrong target.

d. Determine the mean acquisition range for air-to-air and air-to-ground communications.

e. Determine the mean range at which station lock was maintained.

f. Determine the angular displacement of the beam at which communications and/or station lock can be maintained.

g. Determine the maximum range that communications can be conducted without compromise.

h. List each major objective and produce a description of the results.

i. List the range values, that were obtained, in the text.

j. Expected range accuracies are $\pm$ 3% of aircraft altitude as described in paragraph 5.1.1.

5.4.2 Instrumented Flight Test.

a. Perform steps 5.4.1.a through f.

b. Compare the aircraft call times to the TSPI to obtain accurate range information.

c. Performs steps 5.4.1.g and h.

d. Expected range accuracy for this test is $\pm$ 3% of actual range as described in paragraph 5.1.2.

5.5 Subsitute Laser Systems. Utilize the Test Operations Procedures written for the existing system for data reduction, data analysis, and data presentation.

5.6 Effects of Atmospheric Conditions. Atmospheric conditions such as visibility, temperature, relative humidity, rain, fog, clouds, and haze will be evaluated throughout the test for a laser system. Each test objective will address atmospheric conditions as they pertain to that objective.

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5.7 General. The test data from the remainder of the tests will be reduced, analysed, and presented as stated in the appropriate Test Operations Procedures. Specific laser system dependent data will be addressed for each test.

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APPENDIX A

SAMPLE TEST OPERATIONS CHECKLIST

		Date-Initials
1.	Project notebook initiated.	_____
2.	Test plan reviewed and test parameters established.	_____
3.	Test facility and support instrumentation scheduled for test period.	_____
4.	Instrumentation checked for proper functioning and appropriate calibration.	_____
5.	Locks, interlocks and warning devices of test facility checked for proper operation.	_____
6.	Safety requirements accomplished and SOP posted at test facility.	_____
7.	Test personnel briefed on test requirements, special procedures and hazards, and security requirements.	_____
8.	Test item checked for handling and shipping damage.	_____
9.	Test item and instrumentation installed.	_____
10.	Tests performed and data recorded.	_____

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TEST PARTICIPANTS CHECKLIST

MOS _____

Rank _____

Sex _____

Age _____

Height _____

Weight _____

Combat experience _____

Flight time in type aircraft used in testing _____

Total rotary wing flight time _____

Total flight time _____

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APPENDIX B

DATA COLLECTION SHEETS

RANGE SET-UP DATA COLLECTION SHEET

Survey information

Target # 1	Lat	Long
Target # 2	Lat	Long
Target # 3	Lat	Long
Target # 4	Lat	Long

Profile P1

Marker 1	Lat	Long
Marker 2	Lat	Long
Marker 3	Lat	Long
Marker 4	Lat	Long
Marker 5	Lat	Long
Marker 6	Lat	Long
Marker 7	Lat	Long
Marker 8	Lat	Long
Marker 9	Lat	Long
Marker 10	Lat	Long

Profile P2

Marker 1	Lat	Long
Marker 2	Lat	Long
.	.	.
.	.	.
Marker 10	Lat	Long

Profile P3

.
.

Profile P8

Marker 1	Lat	Long
.	.	.
.	.	.
Marker 10	Lat	Long

Figure B-1. Range setup data.

Notes:

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	TARGET # 1	TARGET # 2	TARGET # 3	TARGET # 4
	1	2	3	4
TARGET # 1	x			
TARGET # 2		x		
TARGET # 3			x	
TARGET # 4				x
MARKER # 1				
MARKER # 2				
MARKER # 3				
MARKER # 4				
MARKER # 5				
MARKER # 6				
MARKER # 7				
MARKER # 8				
MARKER # 9				
MARKER # 10				

Profile P1 - Range Calculations Chart

Figure B-2. Sample range calculations chart.

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RANGE READINGS DATA COLLECTION SHEET

Mission No._____Date_____Start Time_____Stop Time_____
Test Engineer_____Pilot_____
Aircraft Type/No._____System Ser #_____
Weather Conditions: Visibility_____Temp._____
Rain_____Fog_____Clouds_____Haze_____
Relative Humidity_____
Target Type:_____
_____

Pass No./Time	Marker #	Target range

Figure B-3. Range readings.

Use back of form for notes and comments.

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MAXIMUM RANGE DATA COLLECTION SHEET

Mission No._____Date_____Start Time_____Stop Time_____
Test Engineer_____Pilot_____
Aircraft Type/No._____System Ser #_____
Weather Conditions: Visibility_____Temp._____
Rain_____Fog_____Clouds_____Haze_____
Relative Humidity_____
Target Type:_____

Pass No./Time	Marker #	First Stable Reading/Time

Figure B-4. Maximum stable range readings.

Notes/Comments:

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RANGE RESOLUTION DATA COLLECTION SHEET

Mission No._____Date_____Start Time_____Stop Time_____
Test Engineer_____Pilot_____
Aircraft Type/No._____System Ser #_____
Weather Conditions: Visibility_____Temp._____
Rain_____Fog_____Clouds_____Haze_____
Relative Humidity_____
Target Type:_____
_____

Rng Vs. Mkr.	Tgt Sep.	Rng to Tgt 1	Rng to Tgt 2

Figure B-5. Range resolution data.

Notes:

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ILLUMINATOR DATA COLLECTION SHEET

Mission No._____Date_____Start Time_____Stop Time_____
Test Engineer_____Pilot_____
Aircraft Type/No._____System Ser #_____
Weather Conditions: Visibility_____Temp._____
Rain_____Fog_____Clouds_____Haze_____
Relative Humidity_____
Target Type:_____
_____

Rng Vs. Mkr.	Time	Range Reading

Figure B-6. Illumination/pinpointing data.

Notes:

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OTHER AIRCRAFT DATA COLLECTION SHEET

Mission No._____Date_____Start Time_____Stop Time_____
Test Engineer_____Pilot_____
Aircraft Type/No._____System Ser #_____
Weather Conditions: Visibility_____Temp._____
Rain_____Fog_____Clouds_____Haze_____
Relative Humidity_____
Target Type:_____
_____

Profile	Rng vs. Mkr.	Time	Range Reading	Altitude

Figure B-7. Illumination/pinpointing data (other aircraft).

Notes:

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COMMUNICATIONS DATA COLLECTION SHEET

Mission No._____Date_____Start Time_____Stop Time_____
Test Engineer_____Pilot_____
Aircraft Type/No._____System Ser #_____
Weather Conditions: Visibility_____Temp._____
Rain_____Fog_____Clouds_____Haze_____
Relative Humidity_____
Target Type:_____

Acquisition Range

Profile	Rng vs. Mkr.	Time	Range Reading	Altitude

Station Lock Range

Profile	Rng vs. Mkr.	Time	Range Reading	Altitude

Figure B-8. Communications data.

Notes:

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APPENDIX C

REFERENCES

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